

13. Positive or negative

Assembly Code:

```

1 LDA 2500
2 ANI 00H
3
4 JZ POSITIVE
5
6 MVI A,01H
7 STA 2501
8 HLT
9
10 POSITIVE: MVI A,00H
11 STA 2501
12
13 HLT

```

Memory Window:

Address (Hex)	Address	Data
09C4	2500	25
09C5	2501	0
09C6	2502	0
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

14. Ascending order

Assembly Code:

```

1 LDA 2500
2 DCR A
3 MOV B,A
4
5 OUTER: LXI H,2501
6 MOV C,B
7
8 INNER: MOV A,M
9 MOV D,A
10 INX H
11 MOV A,M
12 MOV E,A
13 MOV A,D
14 CMP E
15 JNC NOSWAP
16 JZ NOSWAP
17
18 MOV M,D
19 DCR H
20 MOV M,E
21 INX H
22
23 NOSWAP: DCR C
24 JNZ INNER
25 DCR B
26 JNZ OUTER
27
28 HLT

```

Memory Window:

Address (Hex)	Address	Data
09C4	2500	4
09C5	2501	1
09C6	2502	2
09C7	2503	5
09C8	2504	8
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Assembler Message: 0 Program assembled successfully

15. descending order

Assembly Code:

```

1 LDA 2500
2 DCR A
3 MOV B,A
4
5 OUTER: LXI H,2501
6 MOV C,B
7
8 INNER: MOV A,M
9 MOV D,A
10 INX H
11 MOV A,M
12 MOV E,A
13 MOV A,D
14 CMP E
15 JNC NOSWAP
16 JZ NOSWAP
17
18 MOV M,D
19 DCR H
20 MOV M,E
21 INX H
22
23 NOSWAP: DCR C
24 JNZ INNER
25 DCR B
26 JNZ OUTER
27
28 HLT

```

Memory Window:

Address (Hex)	Address	Data
09C4	2500	5
09C5	2501	8
09C6	2502	8
09C7	2503	2
09C8	2504	1
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Assembler Message: 0 Program assembled successfully

16. Largest number

Assembly Code:

```

1 LXI H, 2501
2 MOV B, M
3 MOV A, M
4 MOV D, A
5 DCR B
6
7 LOOP: INX H
8 MOV A, M
9 CMP D
10 JNC SKIP
11 MOV D, A
12
13 SKIP: DCR B
14 JNZ LOOP
15
16 MOV A, D
17 STA 2506
18 HLT

```

Memory Window:

Address (Hex)	Address	Data
09C4	2500	5
09C5	2501	25
09C6	2502	10
09C7	2503	45
09C8	2504	15
09C9	2505	30
09CA	2506	45
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No Assembler Message
0 Program assembled successfully

17. Minimum number

Assembly Code:

```

1 LXI H, 2501
2 MOV B, M
3 MOV A, M
4 MOV D, A
5 DCR B
6
7 LOOP: INX H
8 MOV A, M
9 CMP D
10 JNC SKIP
11 MOV D, A
12
13 SKIP: DCR B
14 JNZ LOOP
15
16 MOV A, D
17 STA 2600
18 HLT

```

Memory Window:

Address (Hex)	Address	Data
09C4	2500	5
09C5	2501	25
09C6	2502	10
09C7	2503	45
09C8	2504	15
09C9	2505	30
09CA	2506	5
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No Assembler Message
0 Program assembled successfully

18. LCM of the given number

Assembly Code:

```

1 LDA 2500
2 MOV B, A
3 LDA 2501
4 MOV C, A
5
6 MOV D, B
7 MOV E, C
8
9 LOOP: MOV A, D
10 CMP E
11 JZ DONE
12 JNC ADDD
13
14 MOV A, E
15 ADD C
16 MOV E, A
17 JMP LOOP
18
19 ADDD: MOV A, D
20 ADD B
21 MOV D, A
22 JMP LOOP
23
24 DONE: MOV A, D
25 STA 2502
26 HLT

```

Memory Window:

Address (Hex)	Address	Data
09C4	2500	6
09C5	2501	8
09C6	2502	24
09C7	2503	0
09C8	2504	0
09C9	2505	0
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No Assembler Message
0 Program assembled successfully

19. GCD of the given number

The screenshot shows an 8086 assembler interface. The assembly code is as follows:

```

02 02 S 0 1 LDA 2500
18 18 Z 1 2 MOV B,A
09 DD AC 0 3 LDA 2501
00 00 4 MOV C,A
42 20 P 1 5
FF FF C 0 6 LOOP: MOV A,B
00 7 CMP C
13 8 JZ DONE
14 9 JNC SECOND
15 10 SUB C
16 11 MOV B,A
17 12 JMP LOOP
18 13
19 14 SECOND: MOV A,C
20 15 SUB B
21 16 MOV C,A
22 17 JMP LOOP
23 18
24 19 DONE: MOV A,B
25 20 STA 2502
26 21 HLT

```

The memory window shows the program starting at address 2500. The assembler message indicates "Program assembled successfully".

20. Factorial of the given number

The screenshot shows an 8086 assembler interface. The assembly code is as follows:

```

10 0 S 0 1 LDA 2500
18 1 Z 1 2 MOV C,A
0D 1B AC 0 3 MVI B,01H
10 4
10 5 FACT: MOV A,C
10 6 CPI 00H
10 7 JZ DONE
10 8
10 9 MOV D,C
10 10 MOV A,B
10 11 MOV E,A
10 12 MVI A,00H
10 13
10 14 MUL: ADD E
10 15 DCR D
10 16 JNZ MUL
10 17
10 18 MOV B,A
10 19 DCR C
10 20 JMP FACT
10 21
10 22 DONE: MOV A,B
10 23 STA 2501
10 24 HLT

```

The memory window shows the program starting at address 2500. The assembler message indicates "Program assembled successfully".